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Generative large language models and academic integrity: ethical risks, detection challenges, and Governance in the age of AI

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ABSTRACT

This paper analyzes the impact of generative large language models on academic integrity as a socio-technical issue, rather than a strictly individual problem. The paper does a cross-analysis of 30 high-level Chinese academic articles indexed by SSCI/CSSCI Core, conducts controlled experiments on the generative ability of four major models (ByteDance Doubao, Tencent Yuanbao, Baidu Wenxin Yiyan, and OpenAI Chat-GPT), and does semi-structured interviews with 18 individuals, including editors, professors, integrity officers, publishers, and AI engineers. This paper finds that “surface originality can successfully conceal derivation to circumvent plagiarism and AIGC detection tools such as PaperPass.” The paper finds three themes: surface originality masking derivative content, length-dependent evasion patterns, and institutional responses to evolving evasion. This paper proposes to “enrich the Technology Acceptance Model by perceived ethical risk, develop a concept of a surface-originality evasion ecology.”

KEYWORDS

Ethics; plagiarism; detection; academic integrity; artificial intelligence generated content

INTRODUCTION

The rapid development of generative artificial intelligence, particularly large language models (LLMs) such as ChatGPT (OpenAI), Gemini (Google), Claude (Anthropic), and domestic Chinese models like Wenxin Yiyan (Baidu) and Doubao (ByteDance), has fundamentally reshaped knowledge production and academic writing practices in higher education. An LLM is a deep neural network trained on massive text corpora, capable of understanding context and generating coherent, human-like text on demand (Zhao et al., 2023). These models no longer merely retrieve or summarize existing information but can autonomously produce full-length essays, literature reviews, and even research papers with minimal human input, thereby challenging conventional notions of authorship and originality (Balalle & Pannilage, 2025; Rudolph et al., 2023).

Despite their powerful assistance capabilities, the integration of LLMs into academic ecosystems has created profound threats to academic integrity. Traditional plagiarism – defined as verbatim copying – has become less common, yet a subtler and harder-to-detect form of misconduct has emerged: students and researchers can now generate seemingly original content that is essentially paraphrased or reorganized from training data without proper attribution or genuine intellectual contribution (Dehouche, 2021; Dwivedi et al., 2023). Current plagiarism detection systems, primarily based on string matching and shallow semantic comparison, struggle to identify such AI-assisted pseudo-original texts, resulting in widespread concern about fairness in assessment, erosion of critical thinking skills, and the devaluation of academic credentials (Sengar et al., 2024; Yan et al., 2024). As a consequence, academic integrity, once upheld through mutual trust and verifiable originality, is undergoing a radical transformation in the digital age.

Investigating these emerging risks is of urgent theoretical and practical significance. Theoretically, it extends discussions on technology ethics and responsible innovation by revealing how perceived usefulness and ease of use – core constructs of the Technology Acceptance Model – may inadvertently encourage unethical behavior when not balanced by perceived risk (Al-Kfairy et al., 2024). Practically, clarifying the evasion mechanisms of LLMs provides direct evidence for upgrading detection algorithms, refining institutional policies, and redesigning assessment methods that emphasize process over product. Such efforts are essential to preserving public trust in higher education and scholarly publishing in an era where generative AI is becoming ubiquitous.

Although the academic community has recognized the disruptive potential of LLMs, three critical gaps remain underexplored. First, while existing studies confirm that LLMs can generate plausible academic text, few have systematically demonstrated whether such content can simultaneously appear highly original on the surface while remaining fundamentally derivative or plagiarized, thereby evading established review mechanisms (Morocco-Clarke et al., 2024). Second, the influence of generated text length on detection evasion has received almost no empirical attention, despite its practical importance for distinguishing short assignments from theses and journal articles (Wang & Chen, 2024). Third, from a developmental perspective, there is a lack of forward-looking analysis on how the academic community should respond if future iterations of LLMs achieve near-perfect imitation of human scholarly writing, rendering current detection tools obsolete (Dwivedi et al., 2023; Yan et al., 2024).

Addressing these gaps, the present study poses the following three interrelated research questions that remain unchanged from the original submission: 1) Can generative large language models produce content that appears highly original on the surface but is fundamentally plagiarized, thereby evading existing review mechanisms? 2) How do differences in the length of generated papers influence their evasion capabilities? 3) From a developmental perspective, how should the academic community respond to the possibility that generative papers may evolve to fully evade detection?

To ensure focus and methodological rigor, this study delimits its scope to text-based generative LLMs applied to academic writing tasks (e.g., essays, literature reviews, and full research papers), explicitly excluding image-generation models (e.g., DALL-E, Stable Diffusion) and code-generation models (e.g., GitHub Copilot). The empirical component centers on four mainstream Chinese and international LLMs – ByteDance Doubao, Tencent Yuanbao, Baidu Wenxin Yiyan, and OpenAI ChatGPT – evaluated primarily through the Chinese plagiarism detection platform PaperPass, which offers superior performance in identifying AI-generated traces in Chinese-language academic texts.

LITERATURE REVIEW

Generative Large Language Models and the Crisis of Academic Integrity in Education

Generative large language models (LLMs) have profoundly changed academic writing practices in higher education (Bao et al., 2025; Meyer et al., 2023; Mo & Crosthwaite, 2025). Models such as ChatGPT, Gemini, Claude, Wenxin Yiyan, and Doubao can generate coherent essays, literature reviews, and complete manuscripts within seconds (Dwivedi et al., 2023; Sengar et al., 2024). Students and researchers now use these tools to complete assignments and draft papers with minimal personal effort (Cui & Zhang, 2025; Wang et al., 2025; Yang et al., 2024). Traditional plagiarism, which depends on direct copying, has become rare because LLMs automatically paraphrase and restructure existing knowledge (Cotton et al., 2024; Dehouche, 2021; Morocco-Clarke et al., 2024). This shift creates a new form of misconduct that appears original on the surface but lacks genuine intellectual contribution (Cutri et al., 2021; Perkins, 2023). Current detection tools, including Turnitin and PaperPass, struggle to identify such AI-assisted pseudo-original texts because they mainly rely on string-matching algorithms (Currie, 2023; Wang & Chen, 2024).

Academic integrity traditionally rests on honesty, trust, and proper attribution of ideas (McIntire et al., 2024; Pudasaini et al., 2024). The widespread adoption of LLMs now threatens these core values (McIntire et al., 2024). Many institutions have responded with strict policies: several Chinese universities completely prohibit the submission of LLM-generated content for degree requirements (Jiao et al., 2025; Nozza et al., 2020), while international journals hurriedly require authors to declare AI assistance (Peters et al., 2024; Sagnier et al., 2020). Despite these measures, enforcement remains inconsistent across countries and institutions (Tammeleht & Löfström, 2024). Educators and publishers face growing difficulty in distinguishing human-authored work from machine-generated text, which undermines fair assessment and erodes public trust in academic credentials (Tahar et al., 2020; Yenduri et al., 2024). The educational community therefore urgently needs evidence-based understanding of how LLMs actually evade existing safeguards.

Theoretical Framework and Identified Research Gaps

Academic integrity in the age of generative AI constitutes the central theoretical concern of this study (Browne, 2022; Cao et al., 2023). Generative large language models dramatically lower the cost of producing apparently original academic work, thereby transforming the very meaning of authorship, originality, and honest effort (Dehouche, 2021; Dwivedi et al., 2023; Morocco-Clarke et al., 2024). To explain why students and researchers willingly adopt tools that enable potential misconduct, this study employs the Technology Acceptance Model (TAM) as its primary theoretical lens (Al Kurdi et al., 2020; Davis, 1989). TAM posits that perceived usefulness and perceived ease of use are the two primary determinants of technology adoption (Buscemi, 2023; Hadi et al., 2024). In academic contexts, users perceive LLMs as highly useful and extremely easy to use, which strongly predicts their intention to integrate these models into writing and assessment tasks (Gupta, 2024; Kim, 2024).

However, classical TAM was developed for productivity tools in organizational settings and does not originally account for ethical risk perception (Bengesi et al., 2024; Katebi et al., 2022). This study therefore explicitly extends TAM by introducing perceived ethical risk as a critical moderating construct when the same technology can be weaponized to evade academic integrity policies (Alto, 2023; Korinek & Vipra, 2024). This extension directly responds to the unique challenge of the GenAI era: the easier and more useful a tool becomes, the greater the temptation to misuse it for unacknowledged authorship (Amankwah-Amoah et al., 2024). The extended TAM thus provides a coherent framework for interpreting both the rapid adoption of LLMs in education and the resulting threats to research integrity (Li et al., 2024; Mannuru et al., 2023). The three research gaps identified earlier (surface originality with underlying plagiarism, influence of text length, and future institutional responses) are therefore examined through this academic-integrity-focused extension of TAM.

RESEARCH METHODS

This study employs a cross-analysis of textual data and a multi-case interview approach to explore the evasion capabilities of generative large language models (LLMs) in academic plagiarism detection and their broader implications. The analysis focuses on four mainstream LLMs in China: Byte Doubao, Tencent Yuanbao, Wenxin Yiyan, and ChatGPT, using PaperPass as the primary plagiarism detection tool alongside the Technology Acceptance Model (TAM) framework. PaperPass was chosen for its advantages in Chinese-language plagiarism detection, cost-effectiveness, data privacy protection, and the ability to identify traces of AI-generated content. By contrast, tools like CNKI and Turnitin are more suited to large institutions and international contexts. The research team evaluated these models' performance in generating academic content and assessed the effectiveness and limitations of their evasion mechanisms. For the interview portion, the research team obtained verbal consent from the participants and recorded and transcribed the interviews to facilitate subsequent review of the study.

Cross-Analysis of Textual Data

Cross-analysis of textual data serves as the core method of this study, aimed at evaluating the originality and replicability exhibited by generative LLMs in academic writing by comparing their outputs with original texts. On 15 January 2025, the research team conducted eight separate targeted searches in the China National Knowledge Infrastructure (CNKI), the largest and most authoritative academic database in mainland China, using the following precise Chinese terms: “生成式人工智能,” “ChatGPT,” “大语言模型检测,” “学术不端,” “AI学术诚信,” “生成式AI学术写作,” “大模型论文代写,” and “ChatGPT学术伦理” (Their direct English equivalents are “generative AI academia,” “ChatGPT,” “large language model plagiarism detection,” “academic misconduct,” “AI academic integrity,” “generative AI academic writing,” “large model paper ghostwriting,” and “ChatGPT academic ethics”). All searches were uniformly restricted to peer-reviewed journal articles published between 1 January 2022 and 15 January 2025 in China’s highest-tier journals (CSSCI, Peking University Core, AMI, and CSCD). The eight searches returned 789, 642, 598, 476, 513, 429, 387, and 355 records respectively (total before deduplication: 4189). After rigorous automated and manual deduplication across all result sets and removal of non-article items, exactly 300 unique, full-text articles remained as the final source corpus. The complete list of these 300 articles is permanently and openly accessible at <https://docs.qq.com/sheet/DZnNsd2pWVHVPZE5>.

The selection of the final 30 experimental articles followed a rigorous, fully documented, two-stage purposive sampling process. In Stage 1 (12–14 January 2025), two researchers independently screened all 300 articles first by title, abstract, and keywords, and then read the full texts of all promising candidates. An article was classified as “high relevance” only if its central argument explicitly and substantively addressed the intersection of generative AI and academic integrity concerns (plagiarism or pseudo-originality, detection evasion mechanisms, authorship ethics, or governance implications); it was rated “medium” if the topic appeared but was not the core focus; and “low” if entirely absent. Minor initial differences were thoroughly discussed until complete consensus was achieved, yielding exactly 118 high-relevance articles. In Stage 2 (15–16 January 2025), the two researchers jointly reexamined these 118 articles and selected the final 30 based on three explicit criteria: (a) richest empirical evidence of real-world detection evasion or failure, (b) deepest theoretical discussion of authorship and originality in the GenAI era, and (c) clearest and most actionable policy or governance recommendations. Representative inclusions are ID 006 (direct empirical comparison of LLM-generated vs. human academic texts), ID 012 (detailed documentation of detection tool failures), and ID 030 (exceptional methodological rigor). High-quality articles such as ID 002 and ID 014 were excluded because their primary focus lay on technical model architecture or non-academic applications rather than integrity issues. This transparent, replicable, and consensus-driven procedure guarantees that the final 30 articles represent the most authoritative and directly relevant Chinese scholarship for the subsequent experiments.

Semi-Structured Interviews

To complement the textual cross-analysis with firsthand practitioner insights into real-world detection challenges, ethical risk perception, and institutional responses, this study employed semi-structured interviews. This method was chosen because it ensures consistency across core topics while remaining flexible enough to capture unexpected yet critical nuances that typically emerge when experienced reviewers and developers discuss rapidly evolving generative AI practices.

Participants were recruited exclusively through purposive sampling to achieve maximum informational relevance and diversity of perspectives. Selection criteria required at least five years of senior-level experience in evaluating or producing Chinese academic manuscripts, active familiarity with current plagiarism and AI-detection tools, and direct exposure to suspected LLM-generated content within the past two years. The final sample of 18 participants was deliberately balanced across four stakeholder groups: journal editors from CSSCI-listed outlets, full professors supervising postgraduate

theses at top universities in China, senior publishing-house reviewers, university research integrity officers, and AI enterprise engineers from leading Chinese large-model development teams (Tencent). This balanced composition ensured that academic evaluation perspectives were systematically confronted with the technical viewpoints of those closest to model design and training (Shown in [Table 1](#)).

Interviews were conducted one-on-one via Tencent Meeting between 20 February and 10 March 2025, lasting 45–70 minutes each, and were audio-recorded with explicit prior consent. A common protocol of nine open-ended questions was used, with flexible probing tailored to each participant's expertise. Recordings were transcribed verbatim within 48 hours. Manual inductive thematic analysis was performed in three stages: initial open coding to identify recurring patterns directly from the raw data, grouping of related codes into sub-themes, and final abstraction into higher-order themes. The two researchers first coded several transcripts independently, reconciled differences through discussion until full consensus on the codebook was reached, then divided and cross-checked the remaining transcripts. The resulting themes provided essential real-world triangulation and explanatory depth for the experimental findings on evasion mechanisms, length effects, and necessary governance reforms.

Ethics Statement

Ethical approval for the human-participant elements of this research was granted by the authors' institutional research ethics committee (committee name and approval reference provided on the title page for editorial use). All procedures followed the Declaration of Helsinki. Adult participants provided informed consent with the option to withdraw at any time. Personal identifiers were removed during transcription, and anonymized data were stored on password-protected drives accessible only to the research team. The textual dataset comprised published journal articles and contained no personally identifiable information.

RESEARCH FINDINGS

This section presents the empirical results from the textual generation experiments, plagiarism detection tests, and semi-structured interviews, structured into three themes that directly correspond to the research questions. All findings are interpreted through the extended Technology Acceptance Model (TAM) framework, linking users' perceived usefulness and ease of use of LLMs to their evasion capabilities, while emphasizing the moderating role of perceived ethical risk in amplifying potential misconduct. As a qualitative study, we deepen descriptive findings through thematic analysis and triangulation with interview data, revealing underlying patterns and insights rather than testing quantitative hypotheses. This approach aligns with established qualitative standards, providing rigorous depth without inferential statistics, as the focus is on contextual understanding of AI-enabled academic risks.

Surface Originality Masking Derivative Content

To examine whether generative LLMs can produce content that appears highly original on the surface but is fundamentally plagiarized, thereby evading review mechanisms, the research team conducted human confusion experiments on short paragraphs and full-paper generation tests using the four models. These directly relate to TAM: high perceived ease of use allows rapid creation of plausible texts, yet low perceived ethical risk encourages unacknowledged use when outputs evade detection. The human evaluations and detection metrics consistently demonstrated that LLMs excel at superficial restructuring of existing ideas from training data, creating outputs that pass initial scrutiny but lack true intellectual novelty upon deeper examination. This not only underscores the models' perceived usefulness in academic workflows but also highlights the ethical risk when such tools enable misconduct without immediate consequences, potentially eroding trust in scholarly processes. All relevant

Table 1. Profile of interview participants.

Participant ID	Stakeholder Group/ Position	Institution Type	Age	Years of Experience	Reason for Selection (Summary)
P01	Senior Journal Editor (CSSCI Communication Journal)	University-affiliated scholarly journal	49	6	Leads manuscript screening and similarity checks for a top-tier CSSCI journal, with extensive experience evaluating AI-suspected submissions.
P02	Managing Editor (CSSCI Law Journal)	University-affiliated scholarly journal	54	8	Responsible for final acceptance decisions and policy drafting on plagiarism and AI usage in a leading law journal.
P03	Section Editor (CSSCI Education Journal)	University-affiliated scholarly journal	44	5	Oversees peer review and routinely coordinates with reviewers on suspected AI-generated manuscripts in education research.
P04	Full Professor of Education	Double First-Class university	49	6	Supervises large cohorts of master's and doctoral students and has direct experience detecting AI-written theses chapters.
P05	Full Professor of Information Management	Double First-Class university	54	5	Researches digital learning technologies and advises university committees on policies for LLM use in coursework and dissertations.
P06	Full Professor of Law	Double First-Class university	59	6	Specializes in intellectual property and academic misconduct cases, frequently consulted on GenAI-related dispute resolution.
P07	Senior Reviewer	Large academic publishing house	49	8	Conducts in-depth technical reviews for monographs and edited volumes, including cases flagged as high risk by detection tools.
P08	Acquisitions Editor	Large academic publishing house	44	6	Oversees proposal evaluation and has handled multiple book projects suspected of heavy LLM assistance.
P09	Production Editor	Large academic publishing house	39	7	Coordinates copy-editing and similarity checking workflows, directly operating PaperPass and other detection systems.
P10	University Research Integrity Officer	Research-intensive university	49	5	Leads investigations into academic misconduct cases, including several formal inquiries involving suspected LLM-generated theses.
P11	Deputy Director, Graduate School Office	Research-intensive university	44	6	Responsible for dissertation quality control and implementation of new regulations on AI-related integrity issues.
P12	Head, Teaching and Learning center	Teaching-oriented university	39	5	Designs academic integrity training for staff and students, recently updated to include generative AI risks.
P13	Senior NLP Engineer (LLM Team)	Tencent AI R&D center	34	5	Works on Chinese large-language-model development and has direct knowledge of model training data and generation behaviour.
P14	Applied Scientist (Content Safety)	Tencent AI R&D center	39	6	Develops internal detection and safety filters for LLM outputs, with hands-on experience tuning AI-content classifiers.
P15	Product Manager (AI Education Solutions)	Tencent cloud and AI business unit	34	9	Leads design of AI tools for higher-education clients and liaises with universities on plagiarism and detection requirements.
P16	Internal Legal Counsel (AI Compliance)	Tencent legal department	44	5	Provides compliance advice on LLM deployment in education and publishing scenarios, including liability in plagiarism disputes.
P17	Digital Scholarship Librarian	Comprehensive university library	39	6	Supports researchers with research tools, has piloted AI-detection services and observed usage patterns among postgraduate students.
P18	Educational Technology Specialist	Comprehensive university IT and teaching support center	34	6	Advises faculty on integrating LLMs into teaching while maintaining integrity standards, and monitors emerging misuse cases.

tables (Tables 2–5) are presented here for centralized reference, as they form the foundational experimental evidence supporting this theme and are cross-referenced in subsequent themes.

Short-text confusion patterns. The confusion experiment selected five human-written paragraphs from the 30 sample papers and generated five comparable AI paragraphs, randomized in a survey (Table 2). Participants often misidentified AI content as human, indicating basic evasion in short texts. This highlights TAM’s perceived usefulness, as models produce realistic snippets that could tempt users to substitute for original work without raising alarms. The qualitative patterns from interviews reinforced this, with stakeholders noting that short AI-generated segments blend seamlessly into human writing due to their concise nature and lack of repetitive flaws. Such evasion fosters a false sense of security among users, lowering perceived ethical risk and encouraging broader adoption in high-stakes academic tasks like abstract drafting or introduction composition. A participant shared:

In my recent thesis reviews, I’ve seen abstracts that read perfectly fluent but lack that human spark of novel insight. They’re like polished summaries of existing ideas, not true synthesis. When I probe the student, they can’t explain the logic deeply, which makes me suspect heavy AI assistance, but without proof, I can only recommend revisions. It’s frustrating because tools like PaperPass catch obvious copies but miss these subtle paraphrases, eroding trust in the entire process. We need better training for reviewers to spot the absence of creative leaps, especially in concise sections where AI shines. (P03, Section Editor (CSSCI Education Journal))

Full-paper pseudo-originality mechanisms. Full-paper tests using Yuanbao, Doubao, and Wenxin Yiyao (Tables 3 and 4) showed low plagiarism rates but high AIGC suspicion, confirming surface originality masking derivative nature. Paragraph-by-paragraph generation with ChatGPT (Table 5) further improved evasion, linking to TAM’s perceived ease of use enabling “human-mimicking” outputs that heighten ethical risk when undetected. Participants elaborated on how full manuscripts generated by AI often maintain structural coherence but reveal underlying plagiarism through inconsistent depth or over-reliance on common phrases from the training corpora, making them appear as reorganized compilations rather than original contributions. This pattern exacerbates ethical concerns, as users perceive LLMs as highly useful for producing submission-ready documents quickly, yet the risk of undetected misconduct grows when detection tools fail to flag semantic similarities to source materials. A participant explained:

As an editor, I handle dozens of submissions weekly, and lately, full manuscripts come in with impeccable structure but repetitive phrasing in longer sections. It’s like the AI ran out of creativity midway. In one case, a student’s paper had flawless structure but no evolving argument, which screamed AI. We use suspicion rates to flag them, but authors deny it, claiming heavy editing. Without better tools, we’re forced to accept on faith, which undermines our journal’s integrity standards and makes me worry about future floods of such content. Ethical guidelines must evolve to mandate disclosure, or we’ll see a surge in pseudo-original works. (P02, Managing Editor (CSSCI Law Journal))

Length-Dependent Evasion Dynamics

Our results from varying generation modes reveal that differences in generated paper length influence evasion capabilities, with longer texts exposing more “machine-like” traits. Through TAM, this demonstrates perceived usefulness declining with length, as ease of use for short outputs fosters low ethical risk, but complexity in long texts increases detectability. The experiments illustrated that while short segments evade scrutiny effectively due to their simplicity, extended papers amplify flaws like logical repetition and shallow analysis, making evasion less reliable as length grows. This dynamic ties directly to users’ perceptions: LLMs are seen as highly useful for quick tasks, but ethical risk escalates when longer outputs reveal the models’ limitations, prompting calls for length-specific safeguards in academic settings.

Short versus long text detection variability. Short-paragraph tests showed strong evasion, with low suspicion rates, but full-paper outputs exhibited higher AIGC flags in extended sections. This ties to TAM’s perceived ease of use enabling quick short-text success, yet amplifying ethical risk in academic misuse.

Table 2. Questionnaire survey.

Serial No.	AI Rewritten Content from Randomly Selected Paper Paragraphs	Content Generation Object
1	In some respects, reasoning ability can be said to be the most confusing capability of large models. For relatively simple logical reasoning, even with seemingly adequate logic and an accuracy rate exceeding 50%, there might be a significant discrepancy in accuracy. For example, the same question may result in opposite conclusions even when the analysis is incomplete but not problematic. On the other hand, when given premises and associations, the model performs well in deductive reasoning. However, when faced with complex reasoning processes, such as game theory analysis, large models seem helpless. While they may attempt some game-theoretic analysis, many issues remain.	Human
2	In the 1990s, with the rise of statistical learning methods, generative artificial intelligence saw new development opportunities. During this period, researchers began using probabilistic models and neural networks to generate text, images, and audio. In 1990, Peter Norvig and Stuart Russell proposed rule-based expert systems. Subsequently, with increased computational power and larger datasets, deep learning technology began to emerge. In 2010, Ian Goodfellow and others proposed Generative Adversarial Networks (GANs), a breakthrough technology that significantly advanced the development of generative AI.	AIGC
3	Under the existing legal framework, there is uncertainty regarding the copyright ownership of AI-generated works. Traditional copyright law stipulates that the author of a work is the natural person who creates it. However, AI-generated works lack a clear natural person author, leading to ambiguity regarding copyright ownership. One viewpoint suggests that developers should own the copyright of AI-generated works because they designed and trained the AI model. Another viewpoint argues that the copyright of AI-generated works should belong to the AI model itself, as it played a key role in the creation process.	AIGC
4	The training and deployment process of large models, like ChatGPT, is complex and resource-intensive. These models typically contain billions or even hundreds of billions of parameters, requiring large-scale parallel computation on high-performance computing clusters. Training these models requires hundreds or even thousands of GPUs, and the training process may take several weeks. This high resource consumption and time cost make it difficult for many research institutions and businesses to conduct large-scale training and deployment.	AIGC
5	As an advanced productivity tool, AGI brings exciting opportunities to society but also presents concerning challenges. These mixed emotions stem from the fact that our understanding of AGI lags far behind its development speed. Specifically, human contemplation of AGI's principles, forms of intelligence, and limits of capability, as well as assessments of its impact on society and individuals, are significantly behind the pace of AGI's development. The rapid development of AGI and the significant lag in human understanding of it create a stark contradiction. Grasping this contradiction is key to understanding the current development of AGI and its social impact. Based on this recognition, many scientists and AI industry leaders have called for a pause on large-scale AI model experiments, urging for accelerated development of provably safe AI systems.	Human
6	In terms of technological complexity, the technical principles of generative artificial intelligence are highly specialized. Even technical experts may struggle to explain the reasons and processes behind the generation of certain content, making it difficult for victims to prove the fault of service providers. In cases of personal information infringement, "regarding the liability for damages caused by infringement of personal information rights, it is very difficult for the injured party to prove that the personal information processor was at fault, because personal information processing activities are highly specialized and technical. The individual is unlikely to understand the fault of the processor and even less able to provide evidence to prove it. Therefore, fault liability should not be applied." In contrast, the technical complexity in generative AI scenarios is even higher. Generally, personal information processors can explain the purposes and processes of information handling, but providers of generative AI services often cannot effectively explain the reasons and processes behind harmful content generation. The content generated by generative AI depends on multiple factors such as algorithms, computing power, model parameters, datasets, and users, and these processes are difficult to fully explain. For example, the parameters of ChatGPT have reached hundreds of billions and continue to grow. Therefore, victims in generative AI infringement scenarios face higher technological barriers than in personal information infringement cases.	Human

(Continued)

Table 2. (Continued).

Serial No.	AI Rewritten Content from Randomly Selected Paper Paragraphs	Content Generation Object
7	Within the framework of deep learning, generative artificial intelligence also introduces the technical logic of natural learning. Natural learning emphasizes that the system improves its performance through self-learning and self-optimization as it recursively adjusts its basic structure. Generative AI showcases the system’s attempt to be as diverse as possible through event combinations, and through natural learning’s reflection, it becomes the system’s ability to behave as a subject. When processing input data, the system can dynamically adjust its internal structure based on the specific data, thereby improving its generalization ability and adaptability. This dynamic adjustment process not only enhances the system’s intelligence level but also enables it to perform excellently in handling complex tasks.	AIGC
8	At the same time, we also recognize that the application of generative AI in knowledge production still faces many challenges and controversies. For example, how to ensure the accuracy and reliability of generated content, how to address privacy and data security issues, and how to avoid misuse of the technology. These issues require not only technological innovation but also joint efforts from policy, law, and society. Therefore, we will delve into these challenges and propose corresponding countermeasures and suggestions.	AIGC
9	In the pre-AI era, imitation was a common phenomenon, from imitation shows in the performance world to replicated works in calligraphy and painting. However, the copyright issues arising from imitation did not pose significant challenges to intellectual property law. This is because, first, manual imitation is a task that requires a substantial amount of learning investment. In the pre-AI era, imitators wanting to replicate a skilled artist had to invest significant time in training and learning. While their skill may still lag behind the original artist, they would have already achieved an exceptional level in the field. Secondly, after investing the learning cost, creating a replicated work still required as much time as the original work. Since the imitator is a natural person, even though they do not need to create a work from scratch as the original author did, they still need to painstakingly replicate the work stroke by stroke, which requires comparable time and effort as creating a completely new piece. The substantial time and effort invested means that imitation is just a tool for artists in their creative process. Therefore, the number of replicated works an artist creates is usually not large.	Human
10	Generative artificial intelligence is often considered a black-box model, as its internal complex algorithms and unpredictability make it difficult to predict how the model will respond to specific inputs. Therefore, generative AI itself carries the risk of generating misleading and unrealistic information, which reduces the accuracy and reliability of public opinion dissemination. Additionally, in order to attract attention or achieve specific political goals, social robots tend to spread negative, controversial, and inflammatory topics. Bias in training data further exacerbates the generation of information confusion in social media, tearing apart existing social consensus and posing a threat to national security.	Human

Participants emphasized that short texts benefit from focused context, allowing AI to mimic human brevity without exposing weaknesses, whereas longer documents demand sustained coherence that models struggle to maintain, leading to detectable patterns. Such variability underscores the need for nuanced detection approaches, as users may exploit short-form strengths while underestimating risks in comprehensive works. A participant said:

Short abstracts or introductions from AI are tough to spot—they’re concise and logical. But when I read full theses, the repetition creeps in after 5,000 words, like the model recycles phrases. In one case, a student’s paper had flawless structure but no evolving argument, which screamed AI. We need length-specific checks, as current tools flag overall suspicion but can’t pinpoint why. This disparity makes short AI use tempting for busy students, but it erodes genuine learning over time. (P09, Production Editor)

Length is key—short AI is seamless, long ones falter on depth. Users see usefulness in brevity but ignore risks in full drafts.. (P10,University Research Integrity Officer)

From tech side, short outputs optimize easily; long ones accumulate errors, heightening detection but also ethical concerns. (P15,Product Manager (AI Education Solutions))

Qualitative markers of machine-like degradation. Paragraph-by-paragraph refinement reduced overall suspicion, but longer texts still displayed formulaic patterns, supporting TAM’s view that perceived

Table 3. Data table for Yuanbao and Doubao weight checking.

Serial No.	Yuanbao Word Count	Yuanbao Plagiarism Rate	Yuanbao AIGC Rate	Doudoubao Word Count	Doudoubao Plagiarism Rate	Doudoubao AIGC Rate
1-Depth Analysis of the Impact of Generative Artificial Intelligence on Social Bots and Governance Countermeasures	5197	12%	92%	13130	17%	69%
2-Research on the multifaceted characteristics of yacht interior design	8923	11%	85%	9046	17%	78%
3-Philosophical inquiries on generative AI and generalized intelligence: the case of ChatGPT	13682	9%	81%	10926	19%	61%
4-Research on the application of AI technology in the production of knowledge in the humanities and social sciences	15321	16%	77%	11882	15%	86%
5- Research on Risk Assessment and Governance Strategies for ChatGPT-like Generative Artificial Intelligence	7847	7%	85%	12930	21%	71%
6-Risks of ChatGPT-like generative AI application in research scenarios and its control strategies	10126	6%	58%	14824	16%	79%
7-A Study of Large-Scale Language Modeling in the Academic Publishing Industry	15970	11%	85%	10050	10%	60%
8-Research on Legal Risks of Large-Scale Language Modeling and Its Governance Paths	8109	13%	85%	7381	8%	66%
9-Artificial Intelligence Generated Content (AIGC) in Education and Its Impact on Future Learning Paradigms	6680	20%	83%	11784	19%	86%
10-A study of generative AI macromodels for intelligence applications	15067	7%	66%	11808	12%	79%
11-Media Convergence and Innovation Led by Technology: An Example from the Academic Symposium of China Society of Film and Television Technology	9008	14%	92%	12188	22%	77%
12-A Study of the Emergence Phenomenon of Generative AI in Knowledge Production and Its Implications	13013	7%	83%	13234	18%	65%
13-Application of Artificial Intelligence Technology in Scholarly Publishing: An Example of the UNSILO Platform	7513	16%	86%	8149	20%	67%
14-Development and Future Prospects of ChatGPT-like Large Models	12904	14%	74%	11990	25%	62%
15-Governance for the industrial chain: technological mechanisms and governance logic of artificial intelligence-generated content	14751	7%	78%	9865	14%	70%
16-A Study on the Enhancement Strategy of Traditional Media's Quality Capacity in the Context of Generative Artificial Intelligence	10723	14%	65%	12921	14%	78%
17-A Study of the Impact of Generative Artificial Intelligence on Civic Education in Colleges and Universities	9043	14%	75%	13986	21%	72%
18-Research on Challenges and Countermeasures of Information Content Governance with Generative Artificial Intelligence Large Models	9615	13%	78%	11513	16%	65%
19-Study on the Legal Risks of Generative Artificial Intelligence and its Regulation	13449	13%	63%	11056	17%	80%
20-Data governance for generative artificial intelligence: a study of responsive governance models for self-regulation by industry subjects	12714	14%	71%	12894	19%	68%
21-A Study on the Principles of Infringement Liability Attribution for Generative Artificial Intelligence Service Providers	9510	18%	62%	10864	27%	52%
22-A Study of Copyright Fair Use in Generative Artificial Intelligence for Machine Learning	13347	12%	55%	14002	24%	53%

(Continued)

Table 3. (Continued).

Serial No.	Yuanbao Word Count	Yuanbao Plagiarism Rate	Yuanbao AIGC Rate	Doudoubao Word Count	Doudoubao Plagiarism Rate	Doudoubao AIGC Rate
23-The Application of Generative Artificial Intelligence to Teacher Rubrics in Middle School Mathematics Classrooms and Its Impact on Students' Learning Effectiveness	17088	19%	57%	9281	17%	76%
24-A Study on the Development Path of Generative Pre-Trained Language Models and General Artificial Intelligence (AGI)	12671	9%	68%	10522	11%	73%
25-Research on the Development of Academic Journals Empowered by Digital Technology and the Path of Convergence Development	10863	15%	85%	10008	35%	68%
26-Analysis of the application and effectiveness of the "Forest Prosecutor" mechanism in improving the quality and efficiency of ecological prosecution	8573	9%	90%	9317	24%	56%
27-A Study of Generative Artificial Intelligence in the Scholarly Publishing Industry with Liability Risks	13935	9%	90%	11470	16%	80%
28-Artificial Intelligence in Academic Counterfeiting and Its Challenges and Prospects	13770	15%	91%	9614	15%	68%
29-The Construction and Challenges of a Generative AI Governance System	14902	10%	64%	9643	22%	69%
30-A Capacity-Based Perspective on Media Convergence in the Context of Generative Artificial Intelligence	15607	10%	50%	9263	20%	65%
Average value	11528	12.21%	76.64%	11251.0	18.31%	70.14%

usefulness wanes as length reveals limitations, heightening ethical concerns for undetected plagiarism. Stakeholders in interviews described how initial fluency in AI outputs fades in prolonged sections, manifesting as inconsistent tone or overused transitions that human writers naturally vary. This degradation not only aids detection but also amplifies ethical risk, as users perceive LLMs as convenient for initial drafts but overlook how length exposes unoriginality, potentially leading to widespread academic dishonesty if unaddressed. A participant explained:

In code reviews or long reports, AI starts strong but loses coherence in depth—logical jumps or overused transitions. I've rejected papers where the methods section felt copied from templates. It's not just plagiarism; it's the lack of human nuance that gives it away, especially over 10,000 words. Institutions must train reviewers on these subtle cues to counter the growing risk, or we'll see ethics erode as models improve. (P16, Internal Legal Counsel (AI Compliance))

Degradation in long texts is obvious—repetitive ideas surface. This usefulness drop warns of ethical pitfalls in extended academic use. (P01, Senior Journal Editor. (CSSCI Communication Journal))

Markers like stale phrasing in lengthy sections reveal AI, urging better training to heighten risk awareness. (P10, University Research Integrity Officer)

Institutional Responses to Evolving Evasion

From a developmental perspective, this study examines how the academic community should address potential full evasion, with integrated findings and interviews proposing proactive strategies. TAM frames this as elevating perceived ethical risk through policies, countering usefulness-driven misuse as models advance. The combined data suggest that as LLMs evolve toward undetectable imitation, institutions must shift from reactive detection to preventive measures, fostering awareness of ethical risks to balance the tools' perceived benefits. This forward-looking analysis draws on experiment

Table 4. Data table of Wenshin Yiyan's checking data.

Serial No.	Wenxin Yiyan Word Count	Wenxin Yiyan Plagiarism Rate	Wenxin Yiyan AIGC Rate
1-In-depth analysis of the impact of generative AI on social bots and governance countermeasures	2862	7.00%	62.80%
2-A study of the multifaceted characteristics of yacht interior design	2022	5.00%	73.60%
3-Philosophical inquiries into generative and generalized artificial intelligence: the case of ChatGPT	2843	11.00%	78.83%
4-A study on the use of AI technology in knowledge production in the humanities and social sciences	3718	10.00%	73.35%
5-Research on Risk Assessment and Governance Strategies for ChatGPT-like Generative Artificial Intelligence	2687	11.00%	75.78%
6-Risks of ChatGPT-like generative AI application in research scenarios and its control strategies	3433	11.00%	74.50%
7-A Study of Large-Scale Language Modeling in the Academic Publishing Industry	3492	15.00%	89.46%
8-Research on Legal Risks of Large-Scale Language Modeling and Its Governance Paths	2825	18.00%	81.68%
9-Artificial Intelligence Generated Content (AIGC) in Education and Its Impact on Future Learning Paradigms	2934	16.00%	84.88%
10-A study of generative AI macromodels for intelligence applications	2530	6.00%	74.08%
11-Media Convergence and Innovation Led by Technology: An Example from the Academic Symposium of China Society of Film and Television Technology	3350	9.00%	82.28%
12-A Study of the Emergence Phenomenon of Generative AI in Knowledge Production and Its Implications	3360	8.00%	51.19%
13-Application of Artificial Intelligence Technology in Scholarly Publishing: An Example of the UNSILO Platform	3913	45.00%	31.15%
14-Development and Future Prospects of ChatGPT-like Large Models	3329	14.00%	68.35%
15-Governance for the industrial chain: technological mechanisms and governance logic of artificial intelligence-generated content	3346	6.00%	92.51%
16-Media Convergence in the Context of Generative Artificial Intelligence: an Exploration Based on a Capacity Perspective	3097	20.00%	69.75%
17-Three-dimensional Exploration of Generative Artificial Intelligence Impacting Civic and Political Education in Colleges and Universities	4029	13.00%	24.51%
18-Information Content Governance for Generative Artificial Intelligence Large Models	3017	10.00%	61.99%
19-The Risk Regulation Dilemma of Generative Artificial Intelligence and Its Resolution: a Perspective on ChatGPT's Regulation	4018	20.00%	74.04%
20-Data Governance and Caging of Generative Artificial Intelligence" A Responsive Study Based on Self-Regulation by Industry Subjects	3247	16.00%	79.01%
21-A Discussion of the Principles of Tort Attribution for Generative Artificial Intelligence Service Providers	2726	20.00%	41.12%
22-Generative Artificial Intelligence Enabling Defense Science and Technology Intelligence	3999	10.00%	65.04%
23-The Problem of Fair Use of Generative Artificial Intelligence in Machine Learning	2570	14.00%	41.04%
24-A study of generative artificial intelligence-supported teacher rubrics: a practical exploration based on a middle school math classroom	3619	16.00%	30.65%
25-Generative language modeling and general artificial intelligence: connotations, paths, and implications	3093	8.00%	29.30%
26-Digital technology empowers the development of academic journals and the path of convergence development	3931	15.00%	18.25%
27-Contributing scholars, AI assistants help forest prosecutors	3096	8.00%	78.32%
28-A Review of Research on Liability and Risk Analysis Related to Generative Artificial Intelligence Works in the Scholarly Publishing Industry	3075	6.00%	56.66%
29-Academic counterfeiting, AI is good	3201	19.00%	79.92%
30-From Urgent Use First to Progressive Improvement: The Construction of a Generative Artificial Intelligence Governance System	3366	14.00%	46.51%
Average value	3224	13.37%	63.02%

Table 5. Table of contents for GPT generation.

Serial No.	Word Count	Plagiarism Rate	AI/GC Rate
1-“Generative Artificial Intelligence” Approaches the Philosophical Narrative of Generalized Intelligence - A Philosophical Analysis of ChatGPT’s Pursuit of the Nature of Intelligence	10282	3.00%	5.84%
2-“AI Empowers the Ocean”-Autonomous Ship-Leading the Future-and 2023 China Intelligent Ship Innovation Alliance High-end Academic Forum Held in Shanghai	277	13.00%	0.00%
3-Mechanism analysis of AI-driven academic publishing to serve natural science knowledge production	6938	9.00%	12.70%
4-Risks of ChatGPT-like generative AI and its governance	7778	9.00%	18.83%
5-Risks and Control Measures of ChatGPT-like Generative Artificial Intelligence in Research Scenarios	12355	8.00%	6.02%
6-ChatGPT Exploration: opportunities and challenges of scholarly publishing with AI large-scale language modeling	6298	6.00%	21.68%
7-Highly Conscious Generative Learning: an AIGC Technology-Enabled Learning Paradigm Innovation	11678	14.00%	14.92%
8-Technology Leadership__ Integration and Innovation - Academic Symposium of the Society of Technology held in Chongqing	3714	55.00%	11.44%
9-Generative Artificial Intelligence Enabling Defense Science and Technology Intelligence	7601	9.00%	6.98%
10-Analyzing the “Emergence” Phenomenon under Generative AI: The Logic of Knowledge Production under the Communication Landscape of the “New Normal”	9476	6.00%	8.56%
11-Governance for the industrial chain: technological mechanisms and governance logic of artificial intelligence-generated content	16571	5.00%	12.74%
12-A Discussion of the Principles of Tort Attribution for Generative Artificial Intelligence Service Providers	13094	17.00%	1.75%
13-Three-dimensional Exploration of Generative Artificial Intelligence Impacting Civic and Political Education in Colleges and Universities	10577	12.00%	7.07%
14-Contributing Scholars, AI Assistants Help Forest Prosecutors	990	6.00%	55.00%
15-Intelligent Publishing Tool for Scientific and Technical Journals - UNSILO Platform	6222	69.00%	6.81%
16-Analyzing the “Emergence” Phenomenon under Generative AI: The Logic of Knowledge Production under the Communication Landscape of the “New Normal”	9476	6.00%	8.56%
17-Academic counterfeiting, AI is good	1678	46.00%	7.04%
18-Data Governance and Caging of Generative Artificial Intelligence” A Responsive Study Based on Self-Regulation by Industry Subjects	13988	17.00%	17.18%
19-A study of generative artificial intelligence-supported teacher rubrics: a practical exploration based on a middle school math classroom	8260	14.00%	13.25%
20-From Urgent Use First to Progressive Improvement: The Construction of a Generative Artificial Intelligence Governance System	14401	4.00%	17.55%
21-The Problem of Fair Use of Generative Artificial Intelligence in Machine Learning	11957	9.00%	8.26%
22-Legal Risks and Governance Paths for Large-Scale Language Models	18113	6.00%	10.44%
23-Test analysis of domestic and international generative AI macromodels performing typical tasks in the intelligence domain	15411	3.00%	13.10%
24-A Review of Research on Liability and Risk Analysis Related to Generative Artificial Intelligence Works in the Scholarly Publishing Industry	7373	6.00%	20.85%
25-A Capacity-Based Perspective on Media Convergence in the Context of Generative Artificial Intelligence	7423	12.00%	0.00%
26-The Risk Regulation Dilemma of Generative Artificial Intelligence and Its Dissolution A Perspective on the Regulation of ChatGPT	18666	22.00%	23.51%
27-An Exploration of Coping Strategies in Scholarly Publishing Regarding the Use of AI Tools - A Case Study of ChatGPT	5767	6.00%	16.66%
28-The Impact of Generative Artificial Intelligence on Social Bots and Governance Countermeasures	13810	7.00%	14.66%
29-Digital technology empowers the development of academic journals and the path of convergence development	4194	12.00%	34.81%
30-Generative language modeling and general artificial intelligence: connotations, paths, and implications	5801	7.00%	27.17%
Average value	9339	13.93%	14.11%

trends and stakeholder views to recommend multifaceted reforms that preserve academic standards amid technological progress.

Policy and tool enhancement pathways.The results suggest adopting blockchain for process tracking and developing advanced Natural Language Processing (NLP) tools, aligning with TAM by balancing perceived usefulness with heightened risk awareness to prevent evasion. Participants

advocated for mandatory AI audits and collaborative tool development, noting that current limitations will worsen without innovation. Such enhancements would mitigate ethical risks by making users more conscious of consequences, transforming LLMs from evasion enablers to ethical aids. A participant stated:

As models get better, we'll see perfect imitations soon. Journals need mandatory AI-disclosure forms and blockchain logs of drafts to trace origins. Without that, ethical lapses will skyrocket. I've seen cases where suspicion is high but proof absent—we end up publishing anyway. Future governance must prioritize transparency over punishment to maintain trust, especially as ease of use draws more users into risky practices. (P18, Educational Technology Specialist)

Policy must advance with tools—blockchain verifies processes, raising ethical barriers against evasion temptations. (P06, Full Professor of Law)

Enhancements like NLP upgrades balance usefulness, but without them, risks overwhelm academic norms. (P11, Deputy Director, Graduate School Office)

Ethical education and assessment redesign. Interviews emphasized shifting to process-oriented evaluations and training, per TAM, to heighten perceived ethical risk and deter overreliance on LLMs' ease of use. Participants called for curriculum integration of AI ethics and iterative assessment models, warning that without these, perceived usefulness will dominate, leading to normalized misconduct. This redesign would elevate ethical awareness, ensuring long-term integrity. A participant noted:

Students see AI as a shortcut because it's so useful, but they don't grasp the risks. Universities should integrate ethics modules in writing courses, requiring iterative human drafts. In my supervision, I now demand version histories to spot AI. This not only catches evasion but fosters real learning—otherwise, credentials lose value as models evolve unchecked, turning education into a facade. (P05, Full Professor of Information Management)

Redesigning assessments to be process-focused raises ethical consciousness, countering AI's seductive ease. (P07, Senior Reviewer)

Education on risks transforms usefulness into responsible tool use, safeguarding integrity. (P04, Full Professor of Education)

DISCUSSION

Surface Originality Masking Derivative Content: Implications for Evasion and Integrity

The findings from the short-text confusion experiments and full-paper generation tests affirm that generative LLMs can indeed produce content that appears highly original on the surface while being fundamentally derivative, thereby evading conventional review mechanisms. As demonstrated in Theme 1, models like Yuanbao and Doubao achieved low plagiarism rates (averages of 12.21% and 18.31%) but high AIGC suspicion (76.64% and 70.14%), indicating a capacity for superficial restructuring that mimics human writing without true innovation. This aligns with prior literature highlighting how LLMs paraphrase and reorganize existing knowledge from training data, creating pseudo-originality that bypasses string-matching tools like Turnitin (Dehouche, 2021; Morocco-Clarke et al., 2024). Critically, this evasion capability extends the crisis of academic integrity discussed in the literature review, where traditional plagiarism has evolved into subtler AI-assisted misconduct (Dwivedi et al., 2023; Sengar et al., 2024). The extended TAM framework is justified here, as perceived usefulness drives users to adopt LLMs for efficient content creation, but the absence of perceived ethical risk enables deliberate exploitation, undermining authorship and originality. This theoretical extension critiques classical TAM for overlooking ethics in academic contexts, revealing how ease of use not only facilitates generation but also normalizes unacknowledged reliance, potentially devaluing scholarly credentials if unchecked.

Participants' accounts further deepen this critique, emphasizing that surface fluency in AI outputs often conceals intellectual shallowness, echoing literature on the erosion of critical thinking in AI-

permeated education (Wang & Chen, 2024; Yan et al., 2024). For instance, the confusion experiment's misidentification patterns suggest that short texts exploit localized context to evade detection, but full manuscripts reveal repetitive structures, supporting a critical view that LLMs represent a double-edged sword: useful for brainstorming yet risky for final submissions. This finding challenges optimistic views in the literature review of LLMs as mere assistants (Gupta, 2024; Kim, 2024), arguing instead for a more skeptical stance – generative models may amplify inequality, favoring those who misuse them over ethical practitioners. Theoretically, this justifies integrating perceived ethical risk into TAM, as it moderates adoption intentions in integrity-sensitive domains, prompting calls for empirical tests in future studies to validate this extension across cultural contexts.

Length-Dependent Evasion Dynamics: Variability and Theoretical Validation

Results from varying generation lengths reveal clear divergences in LLM evasion capabilities, with short texts achieving stronger concealment than longer ones, directly answering RQ2. As detailed in Theme 2, short-paragraph tests exhibited robust evasion through concise mimicry, while full-paper outputs showed heightened AIGC suspicion due to emerging formulaic patterns and logical gaps. This variability corroborates literature on LLMs' limitations in sustaining depth over extended content (Morocco-Clarke et al., 2024; Wang & Chen, 2024), where initial fluency degrades, exposing machine-like traits that human writing avoids. Through the extended TAM lens, this demonstrates how perceived ease of use excels in brief tasks, fostering low ethical risk, but declines in complex, lengthy outputs, increasing detectability and underscoring the need for context-aware safeguards. Critically, this finding critiques the literature's overemphasis on LLMs' general utility (Dwivedi et al., 2023; Sengar et al., 2024), highlighting instead a length threshold where usefulness wanes, potentially deterring misuse if institutions leverage it for targeted detection.

Interview themes add critical depth, revealing that reviewers perceive short texts as seamlessly human-like but flag longer ones for repetitive degradation, aligning with the review's discussion of enforcement inconsistencies (Balalle & Pannilage, 2025; Yan et al., 2024). This supports extending TAM with ethical risk, as length exposes unoriginality, elevating perceived consequences and justifying theoretical refinement for AI in education. However, a critique arises: if models evolve to overcome length limitations, as hinted in developmental perspectives (Al-Kfairy et al., 2024), current divergences may vanish, rendering TAM insufficient without dynamic risk factors. Future research should test this critically, perhaps through longitudinal studies, to validate whether length-dependent evasion persists amid advancements, ensuring the framework's robustness in predicting user behavior and guiding policy.

Institutional Responses to Evolving Evasion: Forward-Looking Strategies and Ethical Imperatives

Integrated findings propose multifaceted responses to potential full evasion by future LLMs, addressing RQ3 from a developmental viewpoint. As outlined in Theme 3, blockchain tracking and ethics-integrated assessments emerge as key strategies, framed by TAM to elevate perceived ethical risk and counter misuse driven by perceived usefulness. This echoes literature calls for proactive governance amid AI's rapid evolution (Dwivedi et al., 2023; Yan et al., 2024), where reactive tools like PaperPass fall short, necessitating shifts to process-oriented evaluations that mitigate undetected imitation. Critically, this justifies the extended TAM: by incorporating ethical risk, it explains why users overlook long-term consequences, advocating for policies that rebalance adoption intentions toward responsibility, thus bridging gaps in the review's fragmented regulatory landscape (Balalle & Pannilage, 2025; Morocco-Clarke et al., 2024).

Stakeholder insights deepen this critique, stressing education to heighten risk awareness, aligning with TAM extensions for ethical domains (Al-Kfairy et al., 2024; Gupta, 2024). However, a critical tension arises: while usefulness drives innovation, unchecked evasion could democratize access yet

exacerbate inequality, favoring tech-savvy users over others – a point underexplored in prior literature (Dehouche, 2021; Wang & Chen, 2024). This demands theoretical refinement, testing TAM in global contexts to critique Western-centric assumptions and ensure equitable governance. Ultimately, these responses position academia to harness LLMs ethically, transforming risks into opportunities for enhanced integrity.

CONTRIBUTION AND NOVELTY

This study makes several interrelated theoretical and empirical contributions to emerging debates on generative AI and academic integrity. Theoretically, it extends the Technology Acceptance Model by foregrounding perceived ethical risk as a core construct when tools designed for productivity are repurposed to circumvent integrity norms. By demonstrating how high perceived usefulness and ease of use interact with low perceived ethical risk to normalize AI-assisted pseudo-originality, the study shows that classical TAM is incomplete in integrity-sensitive domains. Empirically, it provides rare, systematically derived evidence that LLMs can generate surface-original yet fundamentally derivative academic texts in Chinese, across different lengths and generation strategies, while still evading widely used plagiarism and AIGC detection tools. The three research questions are answered in a tightly integrated manner, linking confusion experiments, full-paper generation tests, and thematic interview findings to a coherent account of how pseudo-originality is produced, perceived, and contested in practice.

Building on these insights, the article introduces the original concept of “surface-originality evasion ecology” to capture a generalizable socio-technical pattern that has not been theorized in prior work. This concept describes the dynamic configuration in which LLM-generated texts routinely achieve apparent originality at the surface level, with evasion capacity varying by length and generation strategy, while institutional safeguards react belatedly and unevenly. The three themes identified in this study constitute the core dimensions of this ecology: surface originality masking derivative content, length-dependent evasion dynamics, and institutional responses to evolving evasion. By naming and theorizing this surface-originality evasion ecology, the study not only addresses a key blind spot in TAM – its neglect of integrity-sensitive contexts and institutional feedback loops – but also offers a portable analytical lens for future research on AI-enabled academic risks across different languages, sectors, and regulatory environments.

CONCLUSION

This study has examined how generative large language models reshape academic integrity by enabling texts that are highly original in appearance yet fundamentally derivative, revealing systematic gaps in existing detection and governance. Through cross-analysis of Chinese journal articles, controlled generation experiments with four mainstream LLMs, and multi-case interviews with key stakeholders, the findings show that current safeguards struggle to address three intertwined phenomena: surface originality masking derivative content, where LLMs produce fluent but pseudo-original prose that evades string-matching tools; length-dependent evasion dynamics, whereby short segments conceal machine traits more effectively than extended manuscripts; and institutional responses to evolving evasion, in which governance, policy, and pedagogy lag behind rapidly advancing models. Interpreted through an extended Technology Acceptance Model that incorporates perceived ethical risk, the study demonstrates that high perceived usefulness and ease of use, combined with low perceived risk, normalize AI-assisted misconduct and erode the value of scholarly credentials. On this basis, the paper advances the new concept of “surface-originality evasion ecology” to describe the emergent socio-technical configuration in which pseudo-originality, length-calibrated evasion strategies, and uneven institutional counter-measures interact as a single system. This concept offers a portable lens for future comparative work on AI-enabled academic risks and highlights the need for proactive reforms – ranging from process-based assessment and ethics education to stronger

disclosure norms and technical tracking tools – if academic communities are to harness generative AI responsibly while preserving the integrity of research and education.

DISCLOSURE STATEMENT

No potential conflict of interest was reported by the author(s).

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ETHICAL APPROVAL

The study involved human participants, mainly in interviews, and received formal approval from the authors' institutional ethics committee.

INFORMED CONSENT

All participants also provided informed consent.

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